From Pixels to Intelligence: Choosing Open-Weight Multimodal Models Through Visual Evidence

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Background and User Task

Multimodal large language models combine images and text to answer questions, interpret documents, and reason about visual information. However, numerous model variants, inconsistent evaluation settings, and scattered practical feedback make selection difficult. Our interactive atlas helps researchers and developers understand the ecosystem, explore user concerns, compare two models, and identify task-specific candidates before testing them locally.

User Task. Which open-weight MLLMs should they test for their task, and what evidence supports that choice?

Acquired Datasets

18,255 Hugging Face Repositories

Retained Model Ecosystem Records

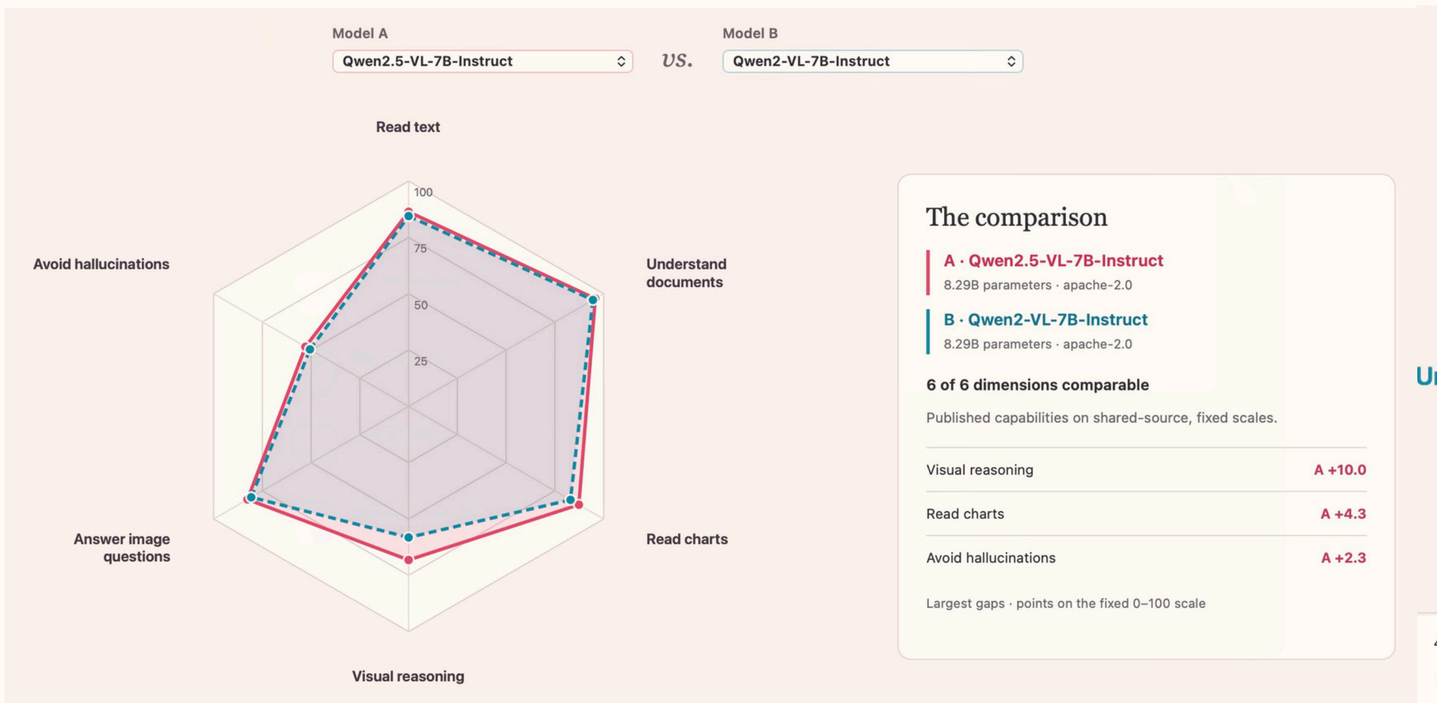
21,674 Community Records

Retained GitHub/Hugging Face Issue Bodies & Comments

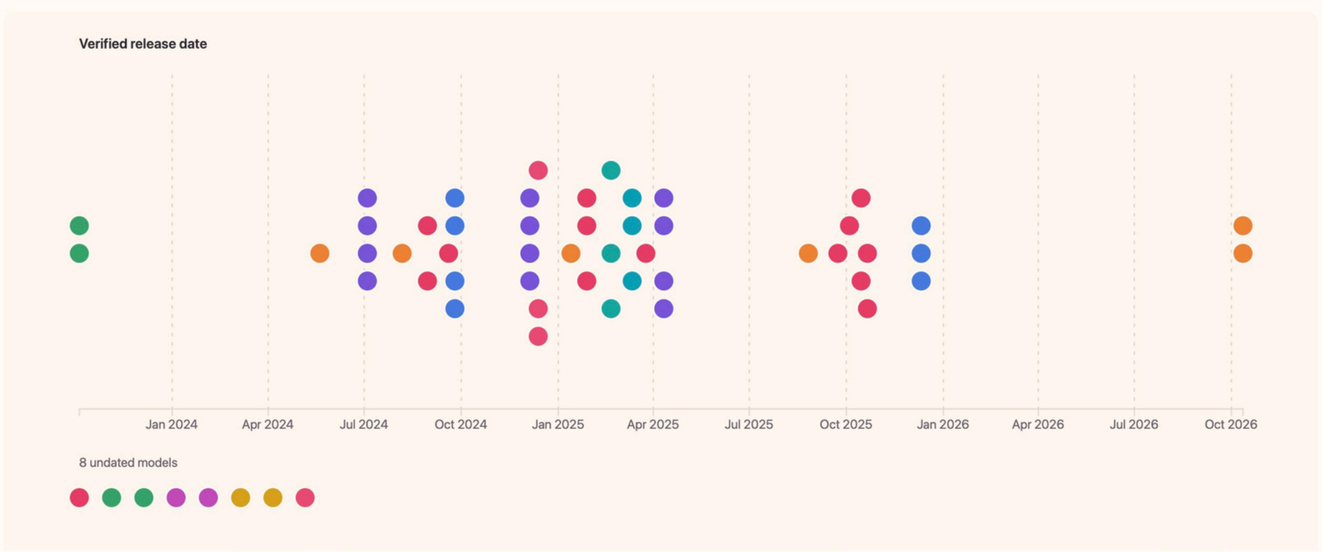
57 Models · 10 Families

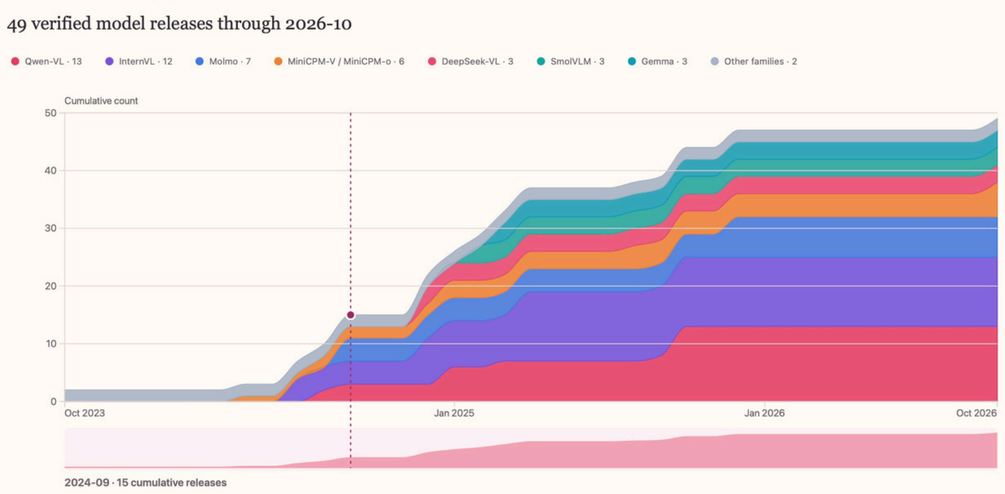
Curated Canonical Comparison Set

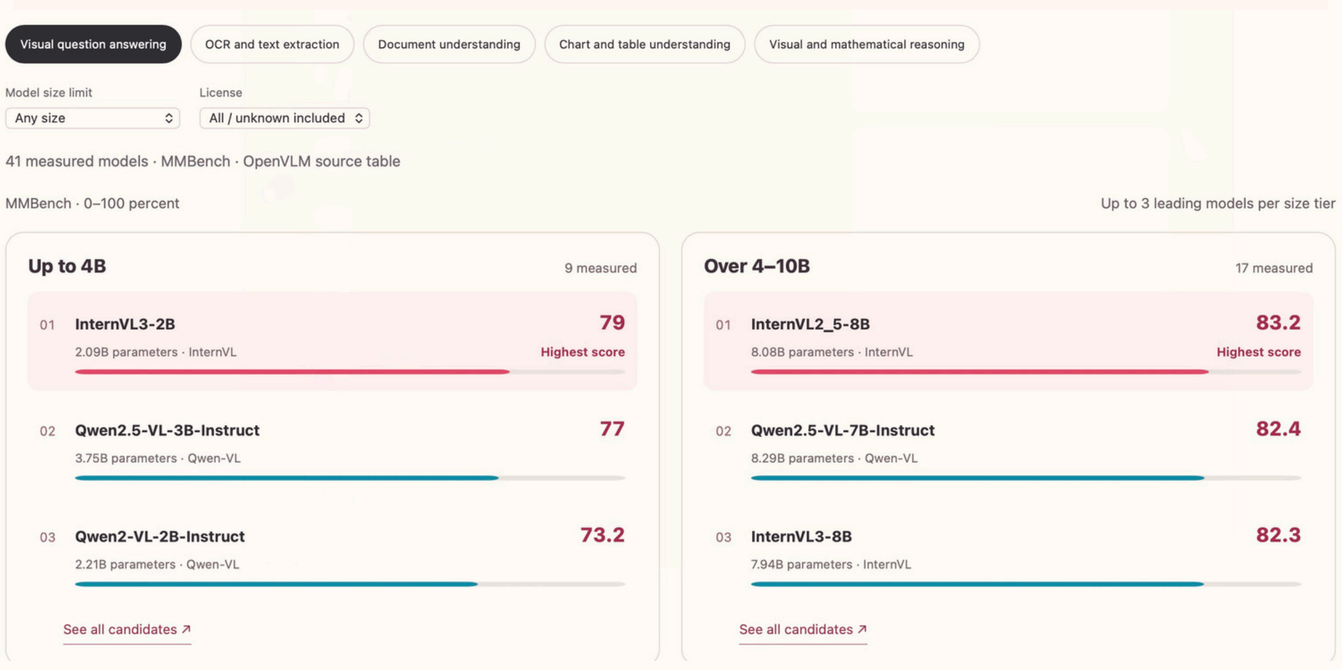
Five Visualizations











From ecosystem overview to task-specific model choice. The five interactive visualizations connect repository growth, community concerns, model-family development, capability comparison, and task-based selection. Users can explore how the ecosystem has expanded, inspect tasks and practical issues mentioned in discussions, follow models across release and family perspectives, compare two benchmark profiles, and identify candidates within parameter-size tiers. Together, these views support evidence-informed model selection while keeping sources, tradeoffs, and missing measurements visible. The goal is to identify suitable models to test, not declare a universal winner.

Core Insights

1. **Repository volume is not original-model diversity.** In the collected snapshot, 10,722 of 18,255 repositories—approximately 58.7%—are classified as quantizations.
2. **Practical concerns extend beyond benchmark accuracy.** Rule-based labeling identifies setup-related mentions in 11,062 records and reliability-related mentions in 6,086. These overlapping categories reflect a support-forum sample, not a representative survey.